Chapter 1

Abelian categories

1.1 Additive categories

Definition 1.1.1 (Additive Category)**.** A Category A is **additive** if it verifies:

A.1) There exists a Zero object $0 \in \mathsf{A}$.

A.2) A has finite Products.

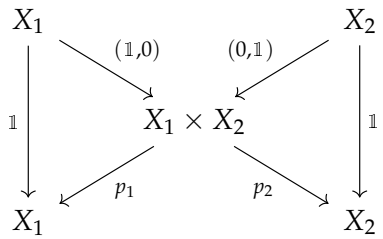
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

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Lemma 1.1.1. *Finite Products and Coproducts coincide in an Additive Category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the Coproduct.

Consider the following diagram



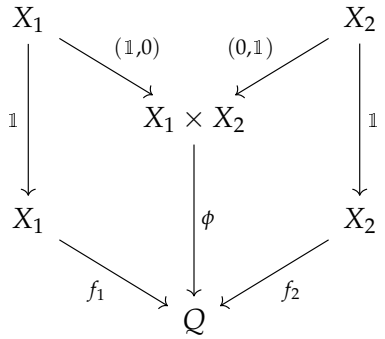
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an Additive Category we will denote its Product (and therefore by Lemma 1.1.1 its Coproduct) by $\bigoplus_{i=1}^n X_i$.

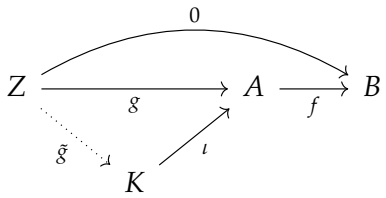
Lemma 1.1.2. *A morphism $\varphi : A \to B$ in an Additive Category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

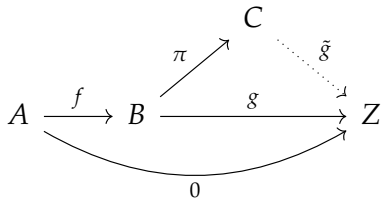
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 1.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



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Definition 1.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

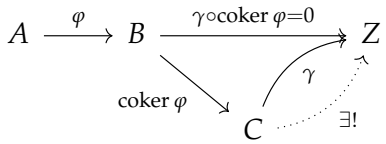


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Lemma 1.1.3. *In an Additive Category Kernels are monomorphisms and Cokernels are epimorphisms.*

Proof. We will only show that Cokernels are epimorphisms. The statement about Kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its Cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the Cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 1.1.2 φ is an epimorphism. □

Lemma 1.1.4. *Let $\varphi : A \to B$ be a morphism in an Additive Category. Then φ is a monomorphism if and only if $0 \to A$ is its Kernel and φ is an epimorphism if and only if $B \to 0$ is its Cokernel.*

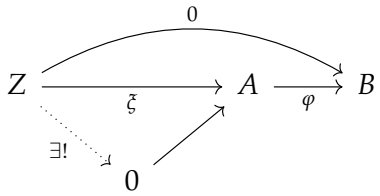
Proof. We will do the Kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 1.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the Kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the Kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the Kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 1.1.2.

□

Definition 1.1.4 (Additive Functor)**.** Let A, B be two Additive Categorys. A Functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

1.2 Abelian categories

Definition 1.2.1 (Abelian Category)**.** An Additive Category A is **abelian** if it verifies

A.4) Kernels and Cokernels exist in A.

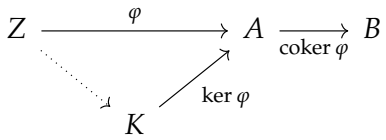
A.5) Every monomorphism is the Kernel of some morphism and every epimorphism is the Cokernel of some morphism.

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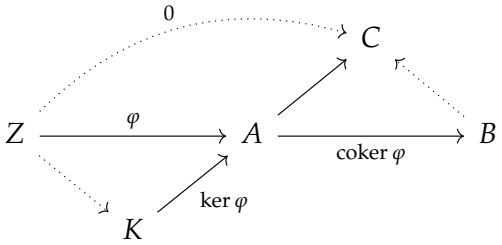
Lemma 1.2.1. *In an Abelian Category* A *every Kernel is the Kernel of its Cokernel and every Cokernel is the Cokernel of its Kernel.*

Proof. Lets check Cokernels this time.

Let coker $\varphi : A \to B$ be the Cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker \varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the Kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the Cokernel universal property we obtain

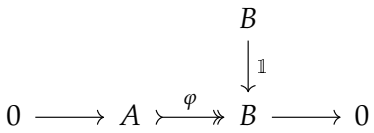


Therefore coker φ satisfies the universal property of the Cokernel of $\ker \varphi : K \to A$. □

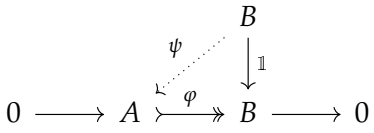
Lemma 1.2.2. *Let $\varphi : A \to B$ be a morphism in an Abelian Category* A. *If φ is both a monomorphism and epimorphism then φ is an Isomorphism.*

Proof. By Lemma 1.1.4 the Kernel of φ is $0 \to A$ and the Cokernel is $B \to 0$. Furthermore by Lemma 1.2.1 φ is the Cokernel of $0 \to A$ and the Kernel of $B \to 0$.

Consider the following diagram



Since $(B \to B \to 0) = 0$ and φ is the Kernel of $B \to 0$ we have



Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram

$$\begin{array}{ccccccccc} & & & & A & & & & \\ & & & & \uparrow{\scriptstyle \mathbb{1}} & & & & \\ 0 & \longrightarrow & A & \overset{\varphi}{\rightarrowtail} & B & \longrightarrow & 0 \end{array}$$

Since $(0 \to A \to A) = 0$ and φ is the Cokernel of $0 \to A$ we have

$$\begin{array}{ccccccccc} & & & & A & & & & \\ & & & & \uparrow{\scriptstyle \mathbb{1}} & \nwarrow{\scriptstyle \eta} & & & \\ 0 & \longrightarrow & A & \overset{\varphi}{\rightarrowtail} & B & \longrightarrow & 0 \end{array}$$

Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

Definition 1.2.2 (Exact sequence)**.** A sequence of morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ in an Abelian Category is **exact** (at Y) if $g \circ f = 0$ and the natural morphism $\operatorname{im} f \to \ker g$ is an Isomorphism.

More generally, a sequence of morphisms $\cdots \to X \to Y \to Z \to \cdots$ is **exact** if every successive pair of arrows is exact. ⌟

Remark 1.2.1. For any morphism $f : X \to Y$ in an Abelian Category we get exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 1.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Definition 1.2.3 (Exact functor)**.** Let A, B be two Abelian Categorys.

A Functor $F : \mathsf{A} \to \mathsf{A}'$ is called left (resp. right) **exact** if for any Exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 1.2.1. Let A be an Abelian Category and let $X \in \mathsf{A}$. The Functors $\operatorname{Hom}_{\mathsf{A}}(X, -)$ and $\operatorname{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

Theorem 1.2.1 (Freyd-Mitchell theorem)**.** *Let* A *be a small Abelian Category. Then there exists a fully faithfull Exact functor* $\mathsf{A} \to R\text{-}\mathsf{Mod}$ *for a suitable (possibly non-commutative) ring* R*, where* $R\text{-}\mathsf{Mod}$ *is the category of left* R*-modules.*